

SRM Institute of Science and Technology

College of Engineering and Technology

School of Computing

SRM Nagar, Kattankulathur – 603203,

Chengalpattu District, Tamil Nadu

Academic Year: 2025-26 (ODD)

Test: CLA1

Course Code & Title: 21IPC501J2 -Research Methodology

Year & Sem: I Year / I Sem

Date: 10-09-2025

Duration: 100 Minutes

Max. Marks: 50

Part – A (25 x 2 = 50 Marks)

Instructions:

1) Answer **ALL** questions.

Question	Marks	BL	CO	PO	PI Code
1. A group of researchers aims to study the impact of Artificial Intelligence (AI)-based tutoring systems on students' critical thinking abilities in higher education. A. Formulate a clear research problem and objectives for the study. (5 Marks) B. Develop an appropriate hypothesis. (5 Marks) C. Suggest a suitable research design (theoretical, experimental, or empirical) and justify your choice. (5 Marks) D. Create a flowchart of the research process you would adopt. (5 Marks) E. Explain how ICT-enabled tools and academic databases can be used to support this research. (5 Marks)	25	4	1	2	1.1.2
[OR]					
2. While conducting a literature review on sustainable smart cities, a scholar finds over 200 papers with overlapping themes. A. Describe how backward and forward searching can refine the review. (5 Marks) B. Explain how grouping and comparative analysis can help synthesize literature meaningfully. (5 Marks) C. Design a strategy to evaluate the quality of the literature using journal-level and author-level metrics (Impact Factor, CiteScore, h-index, g-index, SJR, SNIP). (10 Marks) D. Justify how a meta-analysis could add value in drawing conclusions for this field. (5 Marks)	25	4	1	2	1.2.3
3. A. A researcher has published 25 papers with varying citation counts as shown in the table provided. Using the data, calculate the h-index and g-index of the researcher. Plot a suitable graph to visually represent the indices. (5 Marks)	25	4	1	2	1.3.4

Paper No.	Citations
1	150
2	120

3	95
4	80
5	75
6	65
7	60
8	55
9	50
10	45
11	40
12	35
13	30
14	28
15	25
16	22
17	20
18	18
19	15
20	12
21	10
22	8
23	5
24	3
25	1

B. For Journal "B", Calculate the following bibliometric measures for 2024: Impact Factor and CiteScore.

- Citations received in 2023 = 820
- Citations received in 2022 = 940
- Citations received in 2021 = 700
- Citations received in 2020 = 600
- Papers published in 2023 = 210
- Papers published in 2022 = 190
- Papers published in 2021 = 200
- Papers published in 2020 = 180 (5 Marks)

C. Design a structured questionnaire to study the factors influencing postgraduate students' choice between pursuing research (Ph.D.) and entering the corporate sector.

Include at least:

- 2 Likert scale questions
- 2 Multiple-choice questions
- 2 Ranking/Preference-based questions (5 Marks)

D. Identify the type of research for each of the following cases:

1. A study on the impact of flexible work-from-home policies on employee productivity.
2. Analysis of the yearly rainfall data of a region for the past 50 years.

3. Public opinion survey on the introduction of 5G technology.
4. Investigation of the causes of rising fuel prices.
5. A market study on consumer preference for electric two-wheelers over petrol vehicles.
(5 Marks)

E. A team of researchers plans to evaluate the role of Blockchain technology in securing digital health records.

- Define the research problem.
- State two research objectives.
- Suggest an appropriate research design and justify it. (5 Marks)

[OR]

4. A Ph.D. scholar has developed an innovative machine learning model for early detection of crop diseases.

- A. Identify the possible forms of Intellectual Property Rights (IPR) applicable to the work (Patent, Copyright, GI, Industrial Design, etc.). (5 Marks)
- B. Outline the steps involved in filing a patent in India. (5 Marks)
- C. Discuss how technology transfer can be facilitated between academia and industry for this innovation. (5 Marks)
- D. Evaluate the ethical and societal implications of commercializing this research. (5 Marks)
- E. Suggest strategies to balance open science (knowledge sharing) with protection of innovation. (5 Marks)

25

4

1

2

1.6.1

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Part - A (25 x 2 = 50 Marks)

Instructions:

1) Answer ALL questions.

Question	Marks	BL	CO	PO	PI Code
1. A study aims to explore the influence of creative thinking on problem-solving abilities among corporate employees. A. Define a clear research problem and objectives for this study. (5 Marks) B. Formulate a hypothesis for the relationship between creative thinking and problem-solving skills. (5 Marks) C. Suggest a suitable research design (theoretical, experimental, or empirical) and explain why it fits this study. (5 Marks) D. Draw a flowchart depicting the research process for this study. (5 Marks) E. Describe how ICT-enabled tools and databases can assist in gathering and analyzing relevant data. (5 Marks)	25	4	1	2	1.1.2
[OR]					

2. While reviewing literature on the impact of Artificial Intelligence (AI) in healthcare diagnostics, a researcher encounters a vast number of papers with repetitive findings and varied conclusions.

A. Explain how backward and forward citation searching can help in identifying the most relevant and influential studies.
(5 Marks)

B. Discuss how clustering research based on diagnostic application areas (e.g., radiology, pathology, predictive analytics) and conducting comparative analysis enhances synthesis.
(5 Marks)

C. Propose a systematic method to assess the quality of selected studies using journal-level (Impact Factor, CiteScore, SJR, SNIP) and author-level (h-index, g-index) bibliometric indicators. Include steps and justification.
(10 Marks)

D. Evaluate how a meta-analysis could help in establishing the effectiveness of AI-based diagnostic tools compared to traditional methods across different studies.
(5 Marks)

25 4 1 2 1.3

3. A researcher has published 25 papers with varying citation counts as shown in the table provided. Using the data, calculate the h-index and g-index of the researcher. Plot a suitable graph to visually represent the indices. (5 Marks)

Paper #	Citations
1	120
2	98
3	85
4	76
5	72
6	60
7	48
8	45
9	42
10	39
11	36
12	32
13	29
14	26
15	23
16	21
17	19
18	17
19	14
20	11

25 4 1 2 1.3.4

21	9
22	7
23	5
24	3
25	1

B. For Journal "X," calculate the following bibliometric measures for the year 2025: Impact Factor and CiteScore.

- Citations received in 2024 = 900
 - Citations received in 2023 = 850
 - Citations received in 2022 = 750
 - Citations received in 2021 = 680
 - Papers published in 2024 = 230
 - Papers published in 2023 = 210
 - Papers published in 2022 = 195
 - Papers published in 2021 = 185
- (5 Marks)

C. Construct a structured questionnaire to explore the factors influencing employees' acceptance of remote work, including:

- 2 Likert scale questions
 - 2 Multiple-choice questions
 - 2 Ranking/Preference-based questions
- (5 Marks)

D. Classify the type of research in each of the following scenarios:

- Evaluating the effectiveness of mindfulness training on reducing student stress levels.
 - Studying the changes in average global temperature over the last century.
 - Conducting a survey to understand voter attitudes toward climate change policies.
 - Exploring the reasons behind the decline in newspaper readership in urban areas.
 - Analyzing consumer buying behavior towards organic food products.
- (5 Marks)

E. A group of researchers intends to investigate the impact of cloud computing on improving data storage efficiency in large enterprises.

- Define the research problem.
 - State two research objectives.
 - Suggest an appropriate research design and justify your choice.
- (5 Marks)

[OR]

4. A cybersecurity expert has developed a novel cloud-based encryption algorithm to enhance data security in distributed systems.

25	4	1	2	1.6.1
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A. Identify the possible forms of Intellectual Property Rights (IPR) applicable to this innovation (e.g., Patent, Copyright, Trade Secret, Industrial Design, etc.). (5 Marks)

B. Outline the procedure for filing a patent application in India for this technology. (5 Marks)

C. Discuss methods to facilitate technology transfer of this encryption algorithm from academic research to cybersecurity firms and cloud service providers. (5 Marks)

D. Evaluate the ethical and societal implications of deploying this encryption technology in cloud environments, especially considering privacy and forensic investigations. (5 Marks)

E. Suggest strategies to balance open source collaboration with the protection of intellectual property in cybersecurity innovations. (5 Marks)